

Engineering Catalyst-Electrolyte Microenvironments to Optimize the Activity and Selectivity for Electrochemical Reduction of CO₂ on Cu and Ag

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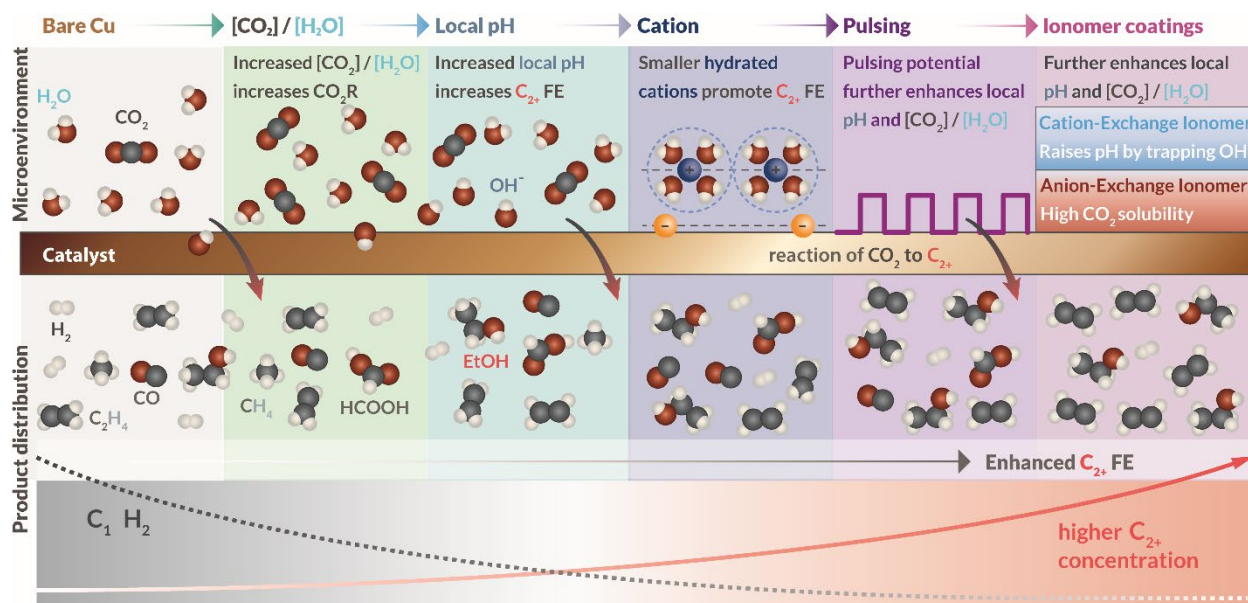
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1. Conspectus

The electrochemical reduction of carbon dioxide (CO₂R) driven by renewably generated electricity (e.g., solar and wind) offers a promising means for reusing the CO₂ released during the production of cement, steel, and aluminum, as well as the production of ammonia and methanol. If CO₂ could be removed from the atmosphere at acceptable costs (i.e., < \$100/t of CO₂), then CO₂R could be used to produce carbon-containing chemicals and fuels in a fully sustainable manner. Economic considerations dictate that CO₂R current densities must be in the range of 0.1 A/cm² to 1 A/cm² and selectivity toward the targeted product is high in order to minimize separation costs. Industrially relevant operating conditions can be achieved by using gas diffusion electrodes (GDEs) to maximize the transport of species to and from the cathode and combining such electrodes with a solid-electrolyte membrane by eliminating the ohmic losses associated with liquid electrolytes. Additionally, high product selectivity can be attained by careful tuning of the microenvironment near the catalyst surface (e.g., the pH, the concentrations of CO₂ and H₂O, and the identity of the cations in the double layer adjacent to the catalyst surface).

We begin this Account with a discussion of our experimental and theoretical work aimed at optimizing catalyst microenvironments for CO₂R. We first examine the effects of catalyst morphology on the production of multicarbon (C₂₊) products over Cu-based catalysts and then explore the role of mass transfer combined with the kinetics of buffer reactions on the local concentration of CO₂ and pH at the catalyst surface. This is followed by a discussion of the dependence of local CO₂ concentration and pH on the dynamics of CO₂R and the formation of specific products. Next, we explore the impact of electrolyte cation identity on the rate of CO₂R and the distribution of products. Subsequently, we look at utilizing pulsed electrolysis to tune the local pH and CO₂ concentration at the catalyst surface. The last part of the discussion demonstrates that ionomer-coated catalysts in combination with pulsed electrolysis can enable the attainment of very high (> 90 %) selectivity to C₂₊ products over Cu in aqueous electrolyte. This part of the Account is then extended to consider the difference in catalyst-nanoparticle microenvironment, present in the catalyst layer of a membrane electrode assembly (MEA), to that of a planar electrode immersed in an aqueous electrolyte.



2. Key References

(1) Singh, M. R.; Clark, E. L.; Bell, A. T. Effects of Electrolyte, Catalyst, and Membrane Composition and Operating Conditions on the Performance of Solar-Driven Electrochemical Reduction of Carbon Dioxide. *Phys. Chem. Chem. Phys.* **2015**, *17*, 18924–18936. <https://doi.org/10.1039/c5cp03283k>.

Explores the impact of mass transport on the electrochemical performance and product selectivity observed in an electrochemical CO₂ reduction device through the means of experiment and macroscale, continuum simulation.¹

(2) Resasco, J.; Chen, L. D.; Clark, E.; Tsai, C.; Hahn, C.; Jaramillo, T. F.; Chan, K.; Bell, A. T. Promoter Effects of Alkali Metal Cations on the Electrochemical Reduction of Carbon Dioxide. *J. Am. Chem. Soc.* **2017**, *139*, 11277–11287. <https://doi.org/10.021/jacs.7b06765>.

Demonstrates the impact of the choice of alkali metal cation on the C₂₊ selectivity of electrochemical CO₂ reduction on Cu catalysts, demonstrating that alkali cations with smaller hydrated radii pack more tightly in the outer Helmholtz plane and stabilize intermediate adsorption.²

(3) Kim, C.; Weng, L. C.; Bell, A. T. Impact of Pulsed Electrochemical Reduction of CO₂ on the Formation of C₂₊ Products over Cu. *ACS Catal.* **2020**, *10*, 12403–12413. <https://doi.org/10.1021/acscatal.0c02915>.

Reveals the impacts of pulsed cathodic potential operation on the observed selectivity and performance of CO₂ reduction on Cu catalysts, demonstrating that pulsed operation enables a substantial enhancement in the selectivity towards C₂₊ products due to an increase in the local CO concentration during dynamic operation.³

(4) Kim, C.; Bui, J. C.; Luo, X.; Cooper, J. K.; Kusoglu, A.; Weber, A. Z.; Bell, A. T. Tailored Catalyst Microenvironments for CO₂ Electroreduction to Multicarbon Products on Copper Using Bilayer Ionomer Coatings. *Nat. Energy* **2021**, *6*, 1026–1034. <https://doi.org/10.1038/s41560-021-00920-8>.

Develops bilayer, ion-conducting polymer coatings for Cu catalysts that carefully tune the local CO₂, water content, and pH to optimize selectivity for C₂₊ products, and provides physical and mechanistic insight for the enhancement of C₂₊ selectivity with these ionomer-coated catalyst materials.⁴

3. Introduction

The rise in the atmospheric concentration of CO₂ has stimulated research devoted to its capture and utilization to produce chemicals and fuels, with the energy provided by renewable sources, such as wind and solar irradiation. If direct air capture of the CO₂ can be done at a cost of < \$100/t, then one could envision a closed-loop cycle for producing chemicals and fuels with no net CO₂ emissions.⁵ Electrochemical reduction of CO₂ (CO₂R) has emerged as a promising means for converting CO₂ into useful products. Technoeconomic analyses have demonstrated that CO₂R current densities in the range of 0.1 to 1 A/cm² and selectivity to multicarbon (C₂₊) products >80% faradaic Efficiency (FE) are required to make CO₂R economically attractive.⁵ Targeted CO₂R current densities and selectivity can be achieved through the use of gas-diffusion electrodes (GDEs) to maximize transport of species and tuning of the chemical microenvironment near the catalyst surface, respectively.^{3,4,6–8}

This Account examines how various factors influencing the microenvironment immediately adjacent to the catalyst surface alter catalyst activity and selectivity for CO₂R. In this context, we examine the effects of mass transfer on the local pH and CO₂ concentration near the catalyst surface, and how the choice of electrolyte cation affects the strength of the electrostatic field in the Helmholtz double layer.^{2,9,10} We show that further control of the microenvironment can be achieved by deposition of thin ionomer layers over the catalyst and by using pulsed, rather than static, electrolysis.⁴ We then turn our attention to the difference in microenvironments in a porous catalyst layer, present in a membrane-electrode assembly (MEA), and planar electrodes immersed in an aqueous electrolyte. Within an MEA, the catalyst nanoparticles in the porous catalyst layer are covered by ionomer, electrolyte, and gases,⁶ creating a complex microenvironment that, at the moment, is poorly understood.

4. Effects of microenvironment on the CO₂R on Cu and Ag

Extensive research by Hori and others, has shown that product selectivity is strongly dependent on the metal used as the cathode.^{11–14} Sn exhibits high selectivity to formic acid and HCOOH, Ag and Au exhibit high selectivity to CO, and only Cu exhibits high selectivity for the formation of multicarbon hydrocarbons and oxygenated compounds.¹⁴ Prior work has also shown that the structure of the catalyst plays an important role in the product distributions observed on various catalysts.^{14–16} In this Account, we focus primarily Cu and Ag catalysts since they can produce C₂₊ products and CO, respectively, with high faradaic efficiency. In the case of Cu, surface roughening to create undercoordinated Cu atoms enhances the formation of C₂₊ products and reduces the formation of H₂ and C₁ products. Since C₂₊ products (e.g., ethylene, ethanol, and propanol) are suitable intermediates for producing a wide variety of chemicals and fuels, substantial efforts have been devoted to identifying methods to alter the microenvironment near the catalyst surface to enhance the C₂₊ product selectivity. In the case of Ag, maximizing the ratio of CO to H₂ can be achieved by increasing the local pH or enhancing the local concentration of CO₂.

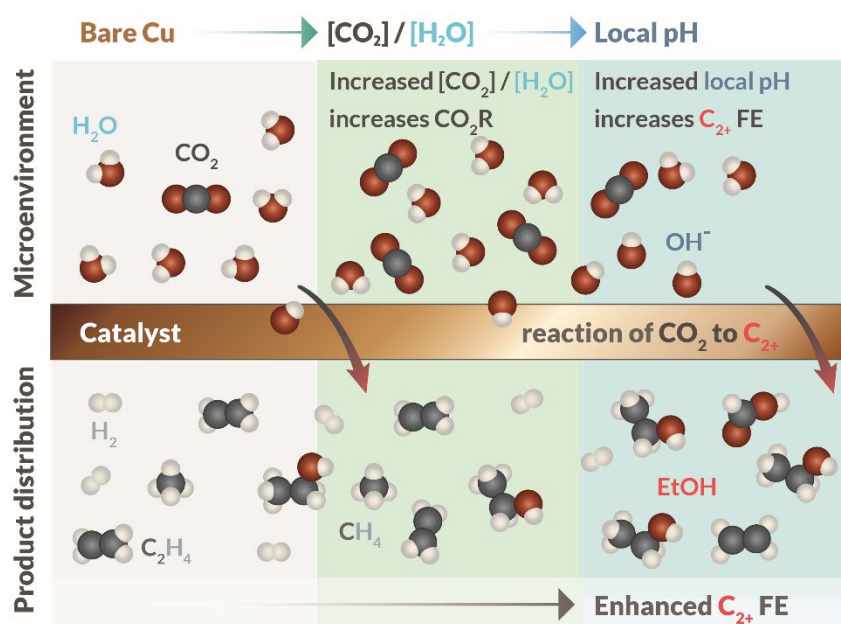


Figure 1: Effects of local microenvironment on CO₂R over a Cu catalyst. Shown are the effects of high local CO₂/H₂O ratio on the rate of CO₂R and the effect of high local pH on the FE to C₂⁺ products.

A schematic showing how the microenvironment near the catalyst surface influences the rate at which CO₂R occurs and its product distribution is given in **Figure 1**. A high ratio of the local concentrations of CO₂ to H₂O enhances the rate of CO₂R, whereas a high local pH suppresses the formation of H₂ and C₁ relative to C₂⁺ products.^{16,17} While not shown in this figure, the microenvironment at the catalyst surface is also influenced by the identity of the electrolyte cations and anions, and by coating the surface with thin films of ionomer.

Lastly, it is critical to note that this introduction to the effects of microenvironment on catalyst performance is done with the aim of giving the reader an understanding of the manner in which the microenvironment can be engineered in order to achieve high activities and selectivities to targeted products of CO₂R. More comprehensive reviews of many of the topics treated in this Account can be found in Refs. 14, and 18-25.^{14,18-25}

4.1 Impact of mass-transfer effects

During CO₂R in an aqueous electrolyte, H₂O is consumed by the reduction of CO₂, and OH⁻ anions are produced, viz., $\text{H}_2\text{O} + \text{CO}_2 + 2 \text{e}^- \rightarrow \text{CO} + 2 \text{OH}^-$. The CO₂ consumed at the cathode must be supplied by mass transfer from the bulk electrolyte and the OH⁻ anion must transport from the cathode surface to the bulk electrolyte. The gradients in CO₂ and OH⁻ created by these processes are sharpened by the occurrence of bulk reactions involving CO₂ and OH⁻ anion, which produce HCO₃⁻ and CO₃²⁻:



As the cathode current density increases, the local concentration of CO₂ decreases and that of OH⁻ increases, resulting in a substantial increase in the local pH. Computational and experimental studies have shown that the local concentration of CO₂ at the catalyst surface approaches 0 mM (as opposed to the 34 mM of saturated CO₂ in an aqueous electrolyte) due to consumption of CO₂ by both CO₂R and parasitic electrolyte buffer reactions (equations 1-3). Conversely, the pH can rise as much as 4 pH units as a result of the stoichiometric generation of OH⁻ anions during CO₂R.^{1,8,26,27}

The effects of mass transport on the surface concentrations of species are critical to consider because the kinetics of each product generated by CO₂R are dependent on the local concentration of CO₂, the pH, and the overpotential for that reaction. The kinetics for each product have been measured and the product current density can be expressed by a Tafel relationship,²⁸

$$i_k = -i_{o,k} \left(\frac{c_{\text{CO}_2}}{c_{\text{ref}}} \right)^{\gamma_{\text{CO}_2,k}} \exp(-\gamma_{\text{pH},k,\text{SHE}} \text{pH}) \exp\left(-\frac{\alpha_{c,k} F}{RT} (\eta_k)\right). \quad (4)$$

In Eq. (4), $i_{o,k}$ is the exchange current density for product k , $\gamma_{\text{CO}_2,k}$ is the CO_2 reaction order, $\gamma_{\text{pH},k}$ is the sensitivity of the current density to pH, and $\alpha_{c,k}$ and η_k are the transfer coefficient and overpotential for product k , respectively.

We note that the Tafel rate equation assumes a single rate-determining step for which all other steps are quasi equilibrated, and it uses semi-empirical lumped parameters to describe the charge transfer in the rate-determining step. Additionally, the effects of the electrical double layer are contained implicitly in the charge-transfer coefficient and exchange current density. Nevertheless, the importance of controlling the local CO_2 and pH becomes clear, since the rate of the CO_2R conversion to each product is directly dependent on these local concentrations. This understanding motivates engineering the chemical microenvironment as a means to achieve high total current densities and the desired product distribution for CO_2R .

4.2 Impact of the electrolyte cation and anion

In aqueous CO_2R , the cations and anions from the electrolytic salt are essential components of the microenvironment near the catalyst surface, where these ions can influence selectivity. We have investigated the effects of alkali metal/bicarbonate (HCO_3^-) electrolytes, Li^+ , Na^+ , K^+ , Rb^+ , and Cs^+ , on CO_2R in order to understand better their impact on product formation over Cu.^{2,9,29} Partial current densities corresponding to the formation of HCOO^- , C_2H_4 , and $\text{C}_2\text{H}_5\text{OH}$ demonstrated an increase with respect to the atomic radius of the cation, $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Rb}^+ < \text{Cs}^+$, while those for H_2 , CO , and CH_4 did not change significantly.^{2,29} Additionally, when a mixture of LiHCO_3 and MHCO_3 ($\text{M} = \text{Na}^+$, K^+ or Cs^+) was used as the electrolyte, the product distribution more closely resembled that obtained in the presence of MHCO_3 even when Li^+ cations were only 10% of the total.² Theoretical calculations revealed that the observed effects of cation identity are attributable to the strength of the electrostatic field in the double layer, which increases with

decreasing radius of the hydrated cation in the outer Helmholtz plane and increasing PZC of the metal catalyst.^{2,29} The increase in the electrostatic field as the hydrated cation decreases stabilizes adsorption of surface intermediates having significant dipole moments (e.g., $^*\text{CO}_2$, $^*\text{OCCO}$), thereby increasing CO_2R activity and C_{2+} selectivity (**Figure 2**).²⁹

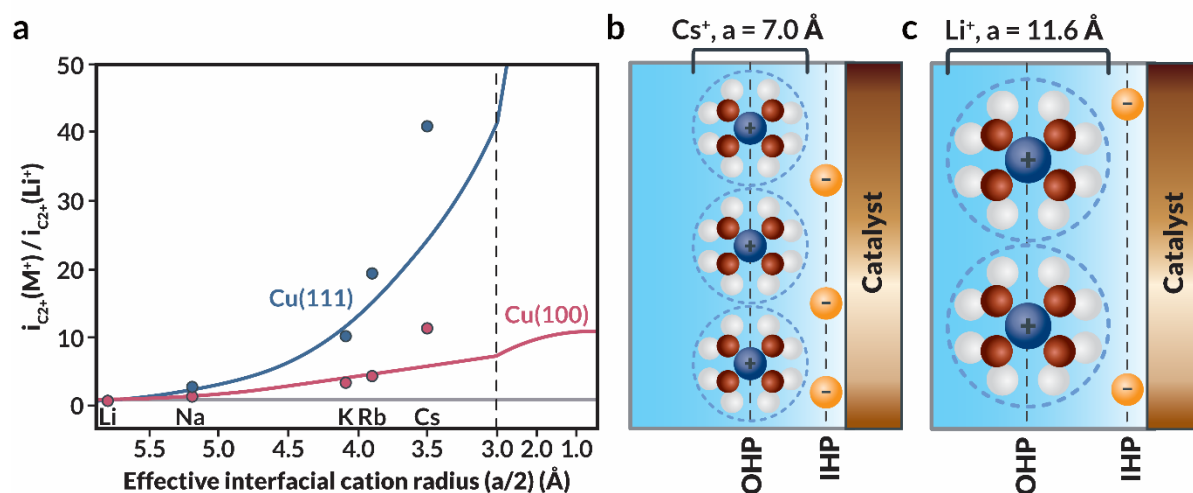


Figure 2: Effect of the electrolyte cation on electrochemical CO_2R on a Cu catalyst.²⁹ (a) C_{2+} current density measured in an MHCO_3 electrolyte normalized by that measured in an LiHCO_3 electrolyte. The blue line is for Cu(111) and the red line is for Cu(100). Points and lines represent the experimental data and theoretical predictions, respectively. (b, c) Illustration of the impact of the size of the hydrated cation on charging of the catalyst surface.

To elucidate the fundamental influence of cations, a multiscale, multiphysics modeling approach is required. Our group has undertaken such an effort for CO_2R on Ag in an aqueous electrolyte. Density functional theory (DFT) was used to predict the rate coefficients needed for the microkinetic model, which was then coupled with a continuum transport model.¹⁰ This approach provided an understanding of the reaction kinetics and surface species present on Ag during CO_2R and revealed that the rate-limiting step for CO_2R on Ag involves hydrogenation of adsorbed CO_2 ($^*\text{CO}_2^{\delta-}$). The studies showed that hydrated cations with smaller radii present near

the surface of Ag stabilize $^*\text{CO}_2^{\delta-}$ through electrostatic interactions,¹⁰ leading to higher coverages and, subsequently, higher rates of CO formation, in agreement with experimental observation.²⁹ Although the complex nature of CO₂R on Cu makes similar studies exceedingly difficult, the findings of this multiscale, multiphysics investigation on Ag demonstrate that cations play a critical role in stabilizing partially charged intermediates, thereby impacting CO₂R product distributions.

We have also investigated the impact of anions on CO₂R over Cu. Anions with buffering capacity, such as bicarbonate, borate (H₃BO₃), and phosphate (HPO₄²⁻), as well as those without buffering capacity, such as perchlorate (ClO₄⁻) and sulfate (SO₄²⁻), were examined at a fixed potassium cation concentration of 0.1 M.³⁰ While anions without buffering capacity exhibited no noticeable influence, buffering anions demonstrated an increase in selective formation of H₂ and CH₄ in the order HCO₃⁻ < H₃BO₃ < HPO₄²⁻. This finding was attributed to the decreasing trend in their pK_a—HCO₃⁻ (10.33) > H₃BO₃ (9.23) > HPO₄²⁻ (7.21), which is consistent with the fact that the rate of H₂ formation is limited by the rate of H atom formation on the catalyst surface via proton-electron transfer, and that the rate of CH₄ formation is limited by the rate of $^*\text{CO}$ hydrogenation.

4.3 Pulsed CO₂ electrolysis

Controlling the cathode potential dynamically is a facile means for modulating the catalyst microenvironment and, therefore, the CO₂R product distribution. Of particular interest is the performance of CO₂R when the cathodic potential is repeatedly alternated between two distinct potential setpoints (i.e., pulsing). Many prior studies of pulsed-CO₂R focused on the pulse-derived reconstruction of the catalyst surface because the cathode potentials were chosen above and below the standard reduction potential of the catalyst (i.e., +0.42 V vs RHE for Cu), resulting in

continuous redox of the catalyst surface.^{31–37} However, improved selectivity toward CO₂R over hydrogen evolution compared to static electrolysis is observed even when both potential setpoints were more negative than the standard reduction potential of the catalyst (see **Figure 3**). This observation implies that pulsed electrolysis can be used to modulate reaction dynamics by perturbing the microenvironment within the vicinity of the catalyst.^{3,36–38}

To investigate the pulsing effect, we employed differential electrochemical mass spectroscopy (DEMS) in tandem with gas chromatography and high-pressure liquid chromatography to analyze both local and bulk concentrations of CO₂R products. DEMS measurements collected *in situ* during CO₂R facilitated direct observation of product-evolution hysteresis between cathodic (more negative potential) and anodic (less negative potential) scans (see **Figure 3a**). Specifically, the generation CO and C₂H₄ was increased in the anodic scan compared to the cathodic scan while that of H₂ exhibited contrary behavior. The two cathodic potential setpoints for pulsed electrolysis were set at –0.8 and –1.15 V vs. RHE to maximize the difference in generation rates between the cathodic and anodic scan, with an optimal duration of 10 s. Relative to static electrolysis at the time-averaged potential, the FE of C₂₊ products increased from 60% to 81% while FE of H₂ decreased from 22% to 9%. DEMS data revealed that the distribution of products generated locally during pulsed electrolysis were distinctly different from those observed for static electrolysis, suggesting differences in the chemical microenvironments between the two electrolysis modes.

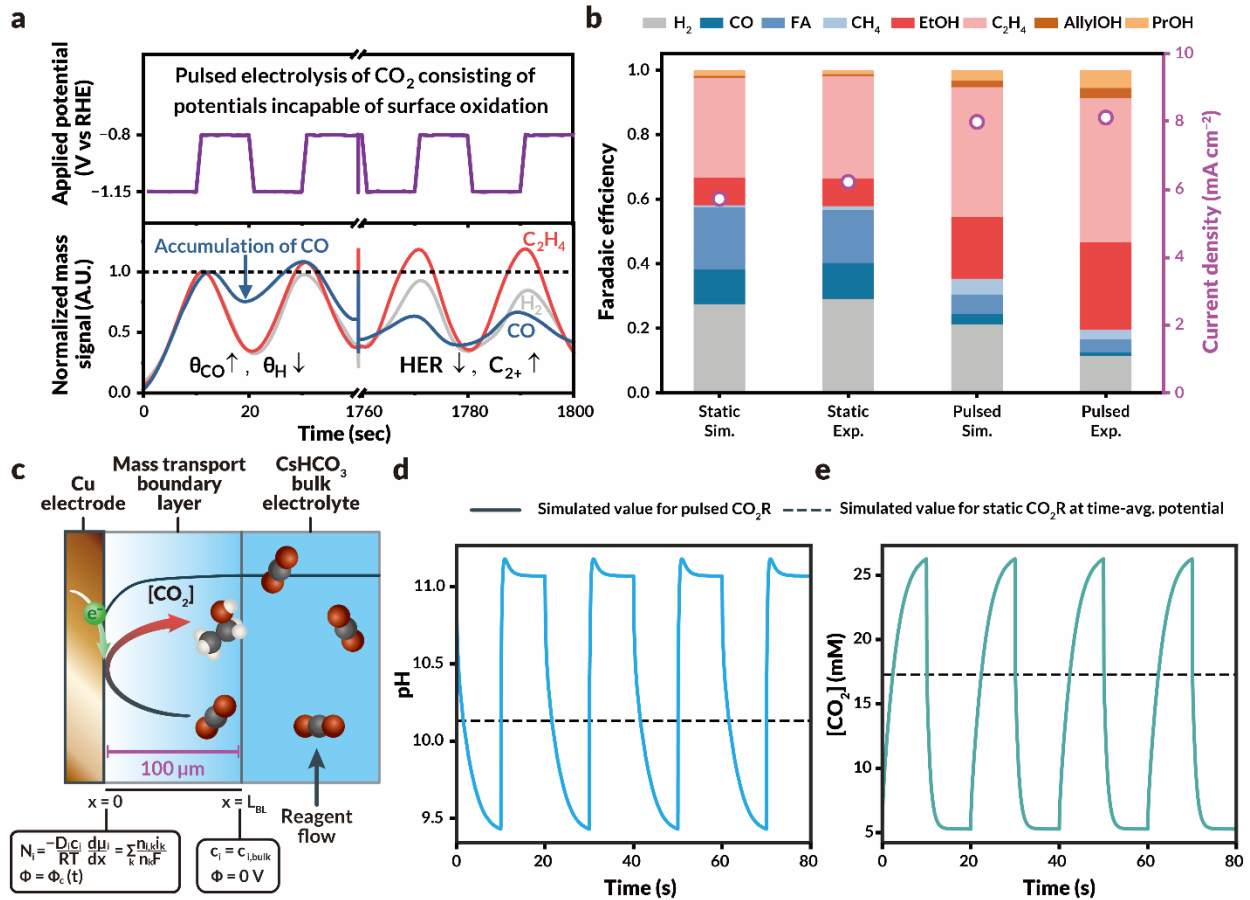


Figure 3: Overview of study of pulsed electrolysis on Cu catalysts.^{3,8} (a) **Top panel:** Rectangular wave applied potential signal used in pulsed electrolysis ($\Phi_1 = -1.15$ V vs. RHE, $\Phi_2 = -0.8$ V vs. RHE, and $t_1 = t_2 = 10$ s). **Bottom panel:** Measured DEMS signal of local C₂H₄, H₂, and CO concentrations at the Cu catalyst surface. (b) Simulated and experimental product distributions and total current density for static and pulsed electrolysis. Static results represent a product distribution taken for static electrolysis at the time-averaged potential. (c) Schematic representation of the simulated domain for the boundary layer model and its boundary conditions. Simulated (d) local pH and (e) local CO₂ concentration for pulsed (solid lines) and static electrolysis (dashed line).

To explain the difference in microenvironments of pulsed and static electrolysis, we developed a time-dependent, continuum model of the mass-transport boundary layer next to the Cu surface (see **Figure 3c**). The electrode surface reactions were modeled using concentration-dependent Tafel kinetics fit to experimental data of static CO₂R on a planar Cu catalyst.⁸ Since the reaction kinetics used in these simulations included the effects of pH and CO₂ concentration, the impacts of mass transport on the local CO₂ concentration and pH and on the observed product

distribution could be described as functions of time. While prior studies attempted to simulate the impact of pulsed electrolysis,^{3,38,39} none explicitly linked the simulated concentrations to product distributions through an experimentally validated kinetic model.

As shown in **Figure 3b**, the model exhibits good agreement with experimentally observed trends; particularly, the enhancements in C_{2+} FE and total current density upon pulsing. Examination of the simulated transient concentration profiles for CO_2 and pH (**Figure 3d-e**) elucidates the cause for the product enhancements observed during pulsed CO_2 R. At the start of the more cathodic portion of the potential pulse, there exists a high transient local concentration of CO_2 , consistent with that observed by DEMS, due to replenishment of the local CO_2 concentration during the less cathodic potential hold. Once the potential is made more cathodic, the local pH rises immediately, due to stoichiometric generation of OH^- .^{6,40,41} At the same time, the local CO_2 concentration decreases due to increased consumption of CO_2 at more cathodic overpotentials. However, because the rate of decrease in local CO_2 concentration is slower than the rate of increase in local pH, a transient state of high local pH and CO_2 concentration is achieved that facilitates attainment of a C_{2+} FE that is far greater than that attainable by static electrolysis. Pulsing the potential enables repeated access to this metastable state and, hence, improved selectivity.

The continuum model of pulsed electrolysis was used to evaluate the performance of pulsed electrolysis for a broad range of possible potentials, as well as in different bulk electrolytes. This effort showed that a shorter potential pulse (~5 s) could be used to further improve C_{2+} FE by enabling more repeated access to the enhanced transient state. Additionally, the model revealed that pulsed electrolysis substantially favors the generation of CH_4 , consistent with experiments when the bicarbonate concentration in the electrolyte was increased because the increased buffer capacity mitigates the increases in pH required to generate C_{2+} products.⁴² These results highlight

the impact of engineering the microenvironment to enable substantial control on the observed product distributions for CO₂R on Cu.

We note that the effects of pulsed electrolysis on the activity and selectivity of Cu have also been observed on Ag. Work by Azumi has demonstrated that on Ag pulsing increases the FE for CO formation and suppresses that for H₂.⁴³ The authors attribute the observed enhancement in CO FE to replenishing of the CO₂ at the catalyst surface during the anodic segment of the pulse (*i.e.*, improved CO₂ mass transport).

4.4 Ionomer coated catalysts

As demonstrated for pulsed CO₂ electrolysis, both high local pH and CO₂ concentration are key to increasing the C₂₊ FE during CO₂R on Cu. However, in an aqueous medium, there is an intrinsic tradeoff between these properties due to the parasitic formation of HCO₃⁻ and CO₃⁼. This tradeoff can be overcome by coating the Cu catalysts with an ion-conducting polymer (ionomer) that tailors the chemical microenvironment to achieve enhanced CO₂R to C₂₊ products.^{44,45} The ionomer composition enables control of ions and water transport through the polymer. **Figure 4** depicts the chemistry and morphology of two standard ionomers: Nafion (a perfluorosulfonic acid, cation-conducting ionomer) and Sustainion (a polystyrene vinylbenzyl methylimidazolium, anion-conducting ionomer).

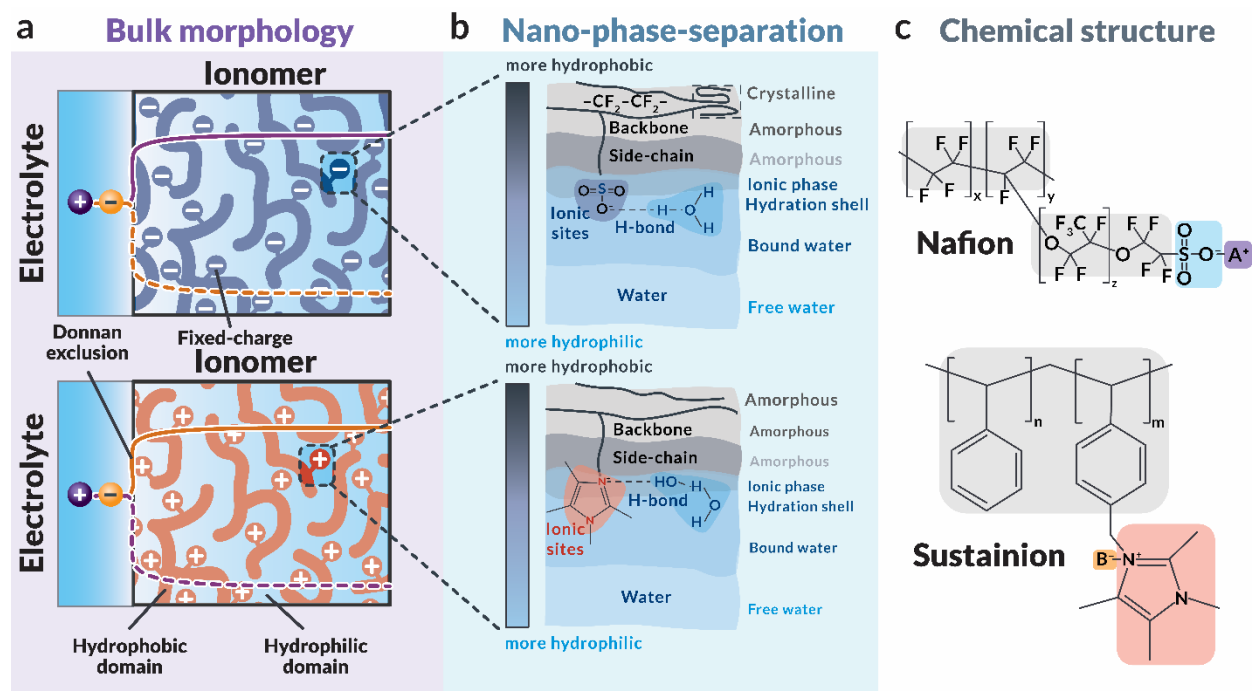


Figure 4: Chemistry and morphology of ionomers. (a) Schematic depiction of the bulk morphology of ionomers. Cation and anion concentration profiles determined by Donnan equilibrium partitioning are shown in the schematic. (b) Phase separation at the nanoscale in the ionomers. (c) Chemical structure of Nafion and Sustainion ionomers.

Our group has demonstrated that a bi-layer of Nafion and Sustainion facilitates both high local pH and CO_2 concentration, leading to 90% FE to C_{2+} products and only 4% FE to H_2 during pulsed CO_2 electrolysis (**Figure 5**).⁴ To understand the role of these ionomer layers, we measured the solubility of water in Nafion (1100 g mol_{eq}⁻¹ of equivalent weight) and Sustainion coated on Cu (Naf1100/Cu and Sus/Cu, respectively). The concentration of water decreased in the order bare Cu (aqueous solution) > Sus/Cu > Naf1100/Cu. Sustainion has about 20-fold higher CO_2 solubility than water due to its imidazolium groups, which have a high CO_2 affinity.⁴⁶ By contrast, Nafion has a CO_2 solubility comparable to water.⁴⁷ Putting these measurements together led to the recognition that the ratio of $\text{CO}_2/\text{H}_2\text{O}$ should increase in the order of bare Cu (aqueous solution) < Naf1100/Cu < Sus/Cu. This order is projected to result in a higher rate of CO_2R compared to HER, which was experimentally confirmed (see **Figure 5**).

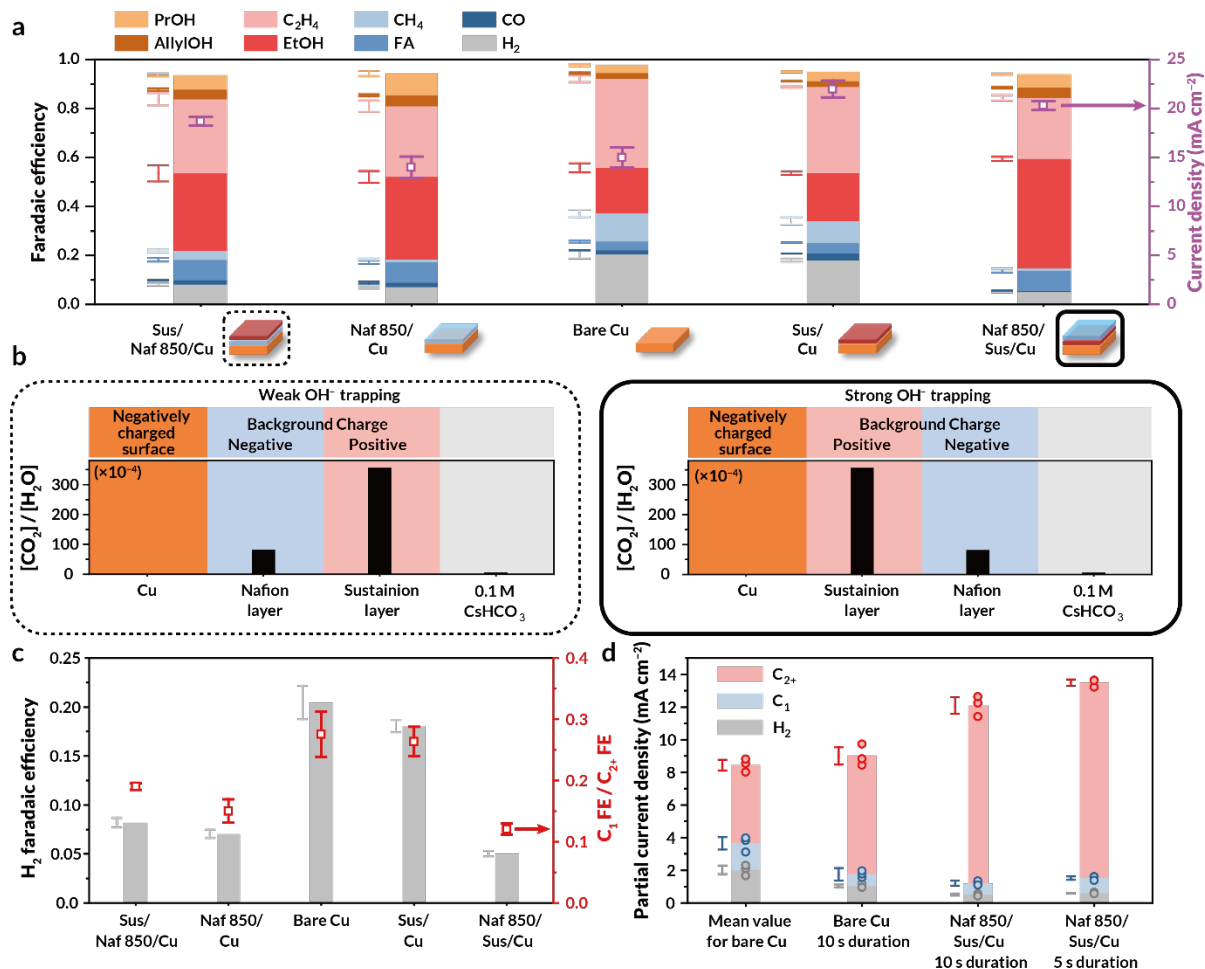


Figure 5: Impact of ionomer layer on CO₂R over Cu catalyst.⁴ (a) Activity and selectivity obtained for different configurations of the ionomer layers. (b) Spatial charge configuration and local [CO₂]/[H₂O] ratio plotted in the various domains of Naf850/Sus/Cu (right) and Sus/Naf850/Cu (left) bilayer ionomer coated catalysts. (c) H₂ selectivity and C₁/C₂₊ FE ratio obtained for different configurations of the ionomer layers. (d) Partial current densities for different configurations of the ionomer employed in pulsed electrolysis. Adapted with permission from Kim *et al.*⁴ Copyright 2021, Springer Science Business Media, LLC.

It is particularly notable that Naf1100/Cu exhibited a higher selectivity to C₂₊ products than bare Cu. Moreover, the C₂₊ selectivity increased further when Naf1100 was replaced with lower equivalent weight Naf850 (850 g mol_{eq}⁻¹), and the *i*_{CO₂R} for Naf850/Cu was lower than that for Naf1100/Cu due to the higher density of hydrophilic sulfonate groups, leading to a lower CO₂/H₂O ratio. Therefore, while the local ratio of CO₂/H₂O explains the enhancement in CO₂R activity, it is not sufficient to describe observed increase in C₂₊ selectivity. Therefore, we postulated that

Donnan exclusion, the thermodynamic exclusion of ions from a polymer matrix of the same sign as that of the background charge (**Figure 4a**),⁴⁸ was responsible for the enhanced C_{2+} selectivity due to an accumulation of OH^- generated at the Cu surface and exclusion by the sulfonate groups in Nafion. This explanation was supported experimentally by observation of a significant decrease in the HER current density when Nafion-coated Cu was used for the electrochemical reduction of acetonitrile, a pH-insensitive reaction that suffers from parasitic HER as a side-reaction.⁴⁹ Finally, as shown in **Figure 5a-c**, by layering Nafion and Sustainion, an optimal microenvironment for C_{2+} production was found to be Naf/Sus/Cu (Nafion at the outermost layer), which achieved a high local CO_2/H_2O ratio due to the inner Sustainion layer and a high local pH due to OH^- trapping by Nafion. This bi-layer architecture resulted in a 250% improvement in C_{2+} production and a 70% suppression in HER during pulsed CO_2 electrolysis relative to what was observed for bare Cu (see **Figure 5d**).

While our studies of ionomer-coated catalysts have been limited to ionomer-coated Cu, similar studies have recently appeared for Ag. Notably, Lees *et al.* invoke a pH trapping argument, similar to that proposed for Cu, in order to explain why Nafion-coated Ag produces substantial amounts of formic acid, a product typically only observed on Ag only at extremely high pH.⁵⁰ And in closely related work, Koshy *et al.* have noted that imidazolium- based ionomer coatings have no effect on CO_2R activity but are able to modulate the activity for HER on Ag.⁵¹ These studies suggest that while some of the ionomer effects we have observed are transferable to Ag, the details of these effects may be chemistry dependent. Hence, further work is needed to explore how ionomers can be used to affect the microenvironment of electrocatalysts used for CO_2R .

5. Porous-electrode systems

A key challenge in the development of CO₂R devices is overcoming mass-transport limitations resulting from the low solubility and diffusivity of CO₂ in aqueous electrolytes.^{1,18,52} Planar electrodes used in conjugation with an aqueous electrolyte are useful for characterizing and/or discovering new catalysts and materials. However, they have relatively thick mass-transfer boundary layers (50-200 μm), which impose mass-transport limitations, limiting CO₂R current density to < 100 mA cm⁻² (**Figure 6a-b**). If CO₂R is to be employed industrially, current densities of > 100 mA cm⁻² must be achieved.⁵ For such applications, devices employing porous electrodes will be necessary (**Figure 6d**). The catalyst layer in a porous electrode is made up of nanoparticles that increase substantially the catalytically active surface area relative to that available for planar electrodes. Furthermore, the porous structure of the catalyst layer enables more direct delivery of gaseous CO₂ through thin, overlapping layers of electrolyte (10-100 nm), which drastically enhances CO₂ transport (**Figure 6e**); such systems have been demonstrated to achieve current densities for CO₂R exceeding 1 A cm⁻².⁵³

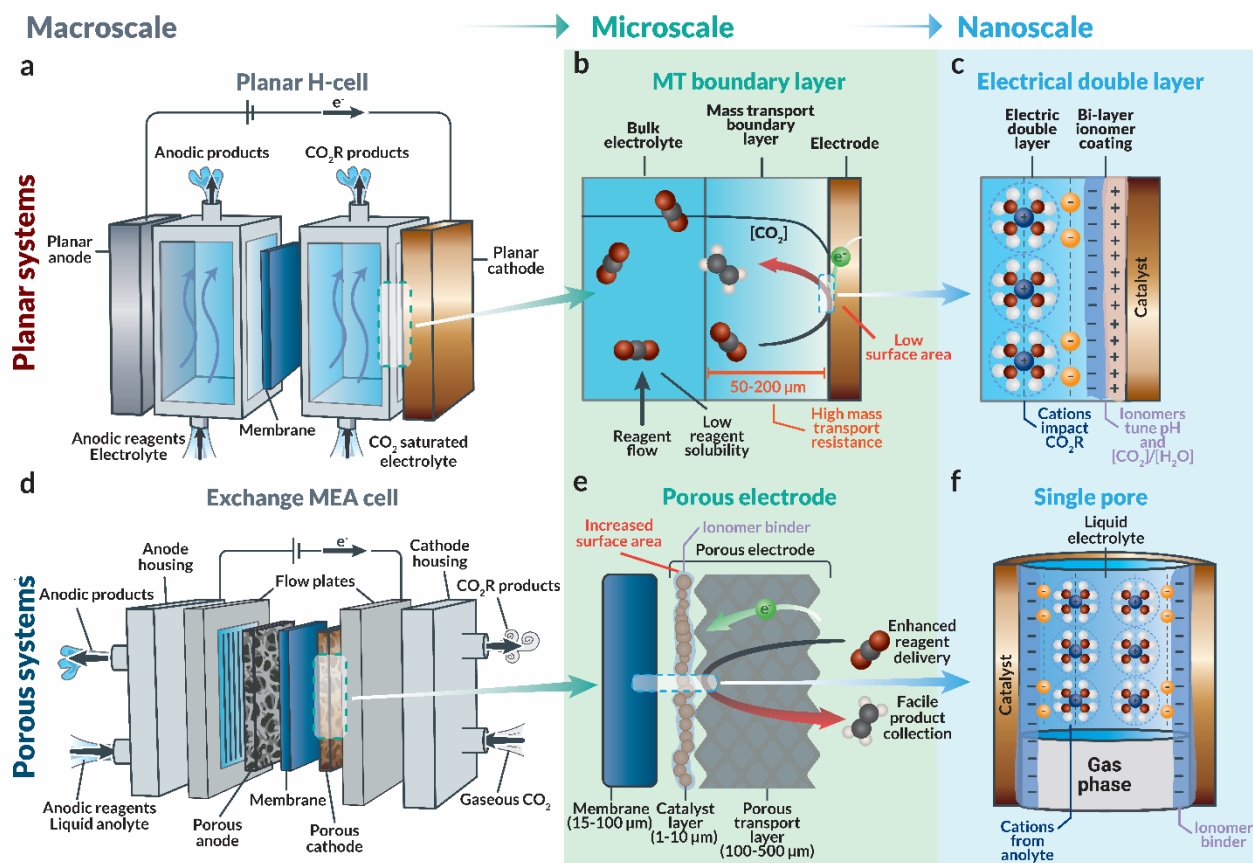


Figure 6: Operation of planar and porous electrode systems at several length scales. (a) Macroscale schematic of a planar H-cell device used for characterizing catalysts for CO_2R . **(b)** Zoomed inset of the mass transport boundary layer in CO_2R on planar Cu. **(c)** Further zoomed inset of the electrical double layer on planar Cu, emphasizing the possibility for cation effects or ionomer layers at the catalyst surface. **(d)** Macroscale schematic of an exchange-MEA device for electrochemical CO_2 reduction. **(e)** Zoomed inset of the porous electrode and the adjacent ion-exchange membrane layer. **(f)** Further zoomed inset of a single pore within the porous electrode.

The most commonly used porous-electrode architecture for CO_2R is the exchange MEA.^{50,53–55} In this architecture, gaseous CO_2 is fed into a flow plate that distributes the reactant CO_2 . The CO_2 diffuses through a tortuous, porous-transport layer (typically a fibrous carbon layer) that ensures uniform delivery of the CO_2 and electrons to the porous catalyst layer. The catalyst layer, consisting of catalyst nanoparticles dispersed in an ionomer binder, is where the CO_2R occurs. The ionomer binds the particles and provides a pathway for ion conduction. An ion-exchange membrane is employed between the porous electrodes to facilitate ionic conduction

between the electrodes, while simultaneously mitigating product crossover. At the anode, an aqueous exchange solution (typically either a hydroxide or a bicarbonate solution) is fed through the porous support for the catalyst layer, wherein water oxidation occurs. The use of an anolyte salt at the anode substantially reduces the ohmic resistance of the overall cell by ensuring the membrane is well hydrated.⁴⁰ **Figure 6** illustrates the operation of planar and porous systems over several length scales. Since this Account is devoted to the effects of the catalyst microenvironment on catalyst performance, the issue we focus on is the difference in microenvironments of a catalyst in a porous electrode to that in a planar electrode with an aqueous electrolyte.

There exist multiple critical differences between the microenvironment in an MEA and that for a planar-electrode system. First, the MEA achieves a much higher total current density compared to that for a planar system due to the substantially reduced diffusion length, which increases CO₂ delivery rate.⁵⁶ However, the high current densities achieved in an MEA produce substantial electro-osmotic fluxes that pull water from the cathode catalyst layer and substantially reduce its water activity.⁶ These fluxes, which do not occur in aqueous systems where the water activity is unity, can drastically skew the ratio of [CO₂]/[H₂O] within the catalyst microenvironment. Lastly, because the diffusion boundary layer in an MEA is quite thin, the effects of pulsed electrolysis, which rely on transient CO₂ and pH changes within the mass-transport boundary layer, may be even greater than those observed for planar electrode systems; hence, future work should evaluate the efficacy of pulsed electrolysis for porous catalyst layers.

We believe that much of the work done by our group on the effects of catalyst microenvironment using planar electrodes is transferable to systems employing porous electrodes. For instance, because it has been shown that cations from the anolyte transport to the cathode catalyst layer during CO₂R in exchange MEAs,⁴⁰ the effects of the electrolyte cation are still

relevant for porous systems.^{54,57} Additionally, the effects of pH and CO₂ availability on the rate of CO₂R are expected to persist in MEAs, as these effects are a result of the intrinsic kinetics of Cu. The selectivity enhancements for ionomer-coated catalysts and pulsed electrolysis were also observed for both metal foils and nanostructured Cu, suggesting that microenvironment effects prevail regardless of the catalyst morphology.^{3,4} Therefore, the effects of the ionomer coatings reported here should be transferrable to porous-catalyst layers comprised of nanoparticles surrounded by an ionomer. For porous catalyst layers supported on the surface of a gas-diffusion layer, the transport of CO₂ to the surface of the ionomer layer surrounding the catalyst is rapid, but its transport to the catalyst surface is dictated by the ionomer, as discussed in recent simulation and experimental studies.^{58–60} Thus we expect the pH effects incurred by Nafion, and the local increase in CO₂ solubility due to the charged N moieties on Sustainion will likely both remain relevant even in porous catalyst layers. Indeed, prior studies have demonstrated that use of a Nafion binder for a Cu catalyst layer drastically enhances the rate of C₂H₄ generation in a porous electrode, and that increasing the Nafion content improves C₂₊ FE.^{53,61} It is important to note, however, that the method of porous-catalyst-layer fabrication typically involves dispersing catalyst particles and ionomer within a solvent and depositing the resulting catalyst-layer ink on the membrane or porous-transport layer. This preparation method poses a significant challenge in the creation of catalyst nanoparticles coated with the kind of distinct bi-layer catalyst coatings developed for the planar system. Thus future work should aim to use techniques such as electrospinning to enable separation of the ionomer binders and facilitate the fabrication of porous catalyst layers with distinct bi-layer ionomer coatings on each nanoparticle.

6. Summary and perspective

This Account demonstrates that not only the composition and surface morphology of the catalyst but also the local microenvironment next to the catalyst surface strongly influence both the catalyst activity and selectivity for the electrochemical reduction of CO₂. We have illustrated how these factors affect both the total current density and the FE for forming C₂₊ products on Cu and CO on Ag. We have shown that a roughened Cu surface enhance exhibits high FE to C₂₊ products and reduced FE to C₁ products and H₂. These effects are largely due to undercoordinated Cu sites on the catalyst surface that bind CO more strongly than do flat facets. Mass-transfer limitations near the catalyst surface, encountered at current densities exceeding about 1 mA/cm², result from a rise in the local pH of the electrolyte and a drop in the local concentration of CO₂ due to the combined effects of mass transfer and buffering via the reaction of CO₂ with OH⁻ anions to form HCO₃⁻ and CO₃²⁻ anions. Electrolyte cations present near the catalyst surface also enhance the activity and selectivity of Cu and Ag catalysts. The choice of electrolyte cation also has an effect on the activity and selectivity of Cu. Cs⁺ cations are present in higher surface concentration in the double layer than cations with a larger radius of hydration, *e.g.*, Li⁺. The higher surface-charge density created by hydrated Cs⁺ cations produces a proportionately stronger space-charge field, which stabilizes the adsorption of CO₂ and the OC-CO dimer that are precursors to C₂₊ products. Because of these effects, the total current density and the C₂₊ FE are higher in CsHCO₃ than LiHCO₃. By contrast, the presence of buffering anions, such as HCO₃⁻ and HPO₄⁻, that have low pK_a's relative to H₂O can serve as proton sources that enhance the formation of CH₄ and H₂ at the expense of C₂₊ products. We have also demonstrated that the placement of a thin ionomer film between the surface of Cu and an aqueous electrolyte further alters the microenvironment near the Cu surface. Most notably, covering the surface with an anion-conducting film, such as Sustainion, and then a cation-conducting film, such as Nafion, enables the attainment of a very

high C_{2+} FE and concurrently a low H_2 FE. These effects are attributed to creation of a microenvironment in which the ratio of CO_2/H_2O and the pH are significantly higher than that in an aqueous electrolyte. Further enhancement of these properties can be achieved by conducting pulsed electrolysis of CO_2 . Finally, we note that the catalyst layer in a membrane-electrode assembly is more complex than that used in a planar-electrode system, opening up additional degrees of freedom for shaping the microenvironment near the catalyst surface.

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